

Ryan Cho

Senior Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Senior full-stack engineer with 7+ years delivering production web platforms across fintech and investment operations. Experienced extending mature Ruby on Rails monoliths, building React and TypeScript interfaces, and optimizing SQL-backed APIs for payment and workflow systems. Pragmatic approach to legacy modernization, background job reliability, and cross-team collaboration in remote environments.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led design and delivery of **payment reconciliation workflows** on a mature Rails monolith, integrating third-party processors and reducing manual operations handling by **40%** across lending pipelines.
- Built and maintained **React and TypeScript** product surfaces for multi-stage approval workflows, improving task completion rates for operations teams managing **complex loan lifecycles**.
- Architected **database schema migrations** and query refactors for high-volume transaction tables, cutting slow API endpoints by **55%** through index optimization and leaner payload design.
- Implemented **Sidekiq-backed background jobs** with retry policies and dead-letter handling for disbursement and notification processing, sustaining **99.7% job success** under peak load.
- Extended legacy Rails services with **versioned REST APIs** consumed by React clients and partner integrations, establishing conventions adopted across **three engineering squads**.
- Partnered with product and compliance stakeholders to scope **incremental feature delivery** on fragile legacy modules, shipping migration paths without service disruption during **18 months** of platform growth.
- Mentored **two junior engineers** on full-stack patterns, SQL profiling, and code review standards, accelerating onboarding and improving pull request throughput across the team.
- Drove reliability improvements through **structured logging, alerting, and runbooks** for payment-critical paths, reducing Sev-2 incidents tied to workflow failures by **30%**.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **internal trading and reporting tools** using Ruby on Rails and React, supporting analysts across global fixed-income desks with **sub-second query response** on daily workflows.
- Optimized **SQL queries and PostgreSQL schemas** for regulatory reporting pipelines, reducing batch runtime by **35%** on datasets exceeding **10 million rows**.
- Built **API integrations** between legacy mainframe exports and modern web dashboards, enabling near-real-time visibility into counterparty exposure for operations teams.
- Maintained and extended **mature Rails applications** under strict change-control processes, delivering features without regression in **critical daily batch cycles**.
- Implemented **background job queues** for document generation and data reconciliation, improving throughput for end-of-day settlement and reporting workflows.
- Collaborated with **business analysts and operations stakeholders** to translate complex financial requirements into scoped technical deliverables across **four cross-functional initiatives**.
- Introduced **TypeScript adoption** on frontend modules within a large monolith, establishing linting and component patterns reused by subsequent feature teams.
- Supported **database migration efforts** during infrastructure upgrades, validating data integrity and rollback procedures across staging and production environments.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **React components** for restaurant inventory dashboards, improving page load performance through lazy loading and component memoization.
- Contributed to **Rails API endpoints** for check-tracking workflows, writing unit tests that increased backend coverage by **20%**.
- Assisted with **SQL query tuning** on PostgreSQL reporting tables, helping reduce dashboard refresh latency for store managers.
- Participated in **Agile sprints and code reviews**, shipping **12+ features** across web and admin tooling during the internship.
- Integrated **third-party payment status webhooks** into internal notification flows, improving visibility for finance operations on settlement delays.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** for a nonprofit fundraising site using HTML, CSS, and JavaScript, supporting a **15% increase** in online donation conversions during the campaign.
- Implemented **CMS-driven content updates** enabling marketing volunteers to publish event pages without developer assistance.
- Optimized front-end assets and image compression, reducing **page load time by 25%** on mobile devices.
- Collaborated with stakeholders to translate **design mockups** into accessible, cross-browser layouts meeting baseline WCAG standards.
- Deployed site updates via **Git-based release workflow**, coordinating launches with campaign timelines and stakeholder review cycles.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, React Native, JavaScript, HTML5, CSS3, Tailwind CSS, Hotwire

Backend: Ruby on Rails, REST APIs, Sidekiq, Background Jobs, Ruby, Node.js

Databases: PostgreSQL, SQL, Schema Design, Query Optimization, Database Migrations, Redis

Cloud & DevOps: AWS, Docker, CI/CD, GitHub Actions, Linux, Heroku

Monitoring & Tools: Sidekiq, Redis, Datadog, Sentry, Git, Jira

Workflow & PM: Agile/Scrum, Technical Documentation, Cross-functional Collaboration, Code Review, Remote Communication